

Local Analytic Rigidity of Isometries on Compact Riemannian Manifolds

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Rigid Rotation on the Circle

For the circle $\mathbb{T} = \mathbb{R}/\mathbb{Z}$, the rotation $R_\alpha : x \mapsto x + \alpha$, $\alpha \in \mathbb{T}$, is **C^r -rigid**, $r \in \mathbb{N}^* \cup \{\infty, \omega\}$, if for any $f : \mathbb{T} \rightarrow \mathbb{R}$ of class C^r , the circle diffeomorphism $F : x \mapsto x + \alpha + f(x)$ is **conjugated to R_α** by a **C^r** diffeomorphism h , i.e.,

$$F = h \circ R_\alpha \circ h^{-1} \quad (F \stackrel{h}{\sim} R_\alpha \text{ or } F \sim R_\alpha)$$

under suitable assumptions on α and F .

- **Local C^r -rigidity** of R_α :

$F \sim R_\alpha$ under the assumption that F is close to R_α (in the sense of C^r)

- **Global C^r -rigidity** of R_α :

$F \sim R_\alpha$ without the closeness assumption

Arithmetic Conditions of Crucial Quantities

- **Angle of rotation** α
 - ▶ **Diophantine** condition: there exist $c, \tau > 0$ s.t.

$$\inf_{l \in \mathbb{Z}} |n\alpha - l| \geq \frac{c}{|n|^\tau}, \quad \forall n \in \mathbb{Z} \setminus \{0\}.$$

- ▶ **Brjuno** condition: for the convergents $\{\frac{p_n}{q_n}\}$ of α , $\sum_{n=1}^{\infty} \frac{\ln q_{n+1}}{q_n} < \infty$.
- ▶ **Herman** condition
- **Rotation number** of circle diffeomorphism F (**Poincaré**): the uniform limit (existing and independent of $x \in \mathbb{R}$):

$$\lim_{n \rightarrow \infty} \frac{\tilde{F}^n(x) - x}{n} \bmod \mathbb{Z} := \rho(F),$$

where $\tilde{F} \circ \Pi$ satisfies $F \circ \Pi = \Pi \circ \tilde{F}$, and $\Pi(x) = x \bmod \mathbb{Z}$

Known Results – Rigidity of Rotation

Linearization of a single circle diffeomorphism F

► **Arnold 1961:**

$F \in C^\omega(\mathbb{T})$, $\rho(F) = \alpha$ **Diophantine** $\implies$ Local C^ω linearization

► **Herman 1979:**

$F \in C^\infty(\mathbb{T})$, $\rho(F) = \alpha$ **Diophantine** $\implies$ Local C^∞ linearization

► **Herman 1979, Yoccoz 1984**

(see also **Khanin-Sinai 1987, Katznelson-Ornstein 1989**):

$F \in C^\infty(\mathbb{T})$, $\rho(F) = \alpha$ **Diophantine** $\implies$ Global C^∞ linearization

► **Yoccoz 1998:**

$F \in C^\omega(\mathbb{T})$, $\rho(F) = \alpha$ **Brjuno** $\implies$ Local C^ω linearization

$F \in C^\omega(\mathbb{T})$, $\rho(F) = \alpha$ **Herman** $\implies$ Global C^ω linearization

Known Results – Rigidity of Group Action by Rotations

Simultaneous linearization of a family of circle diffeomorphisms $\{F_j\}$:
there exists one h s.t. $F_j \stackrel{h}{\sim} R_{\alpha_j}$

► **Moser 1990**:

Finitely many commuting $F_j \in C^\infty(\mathbb{T})$, $\rho(F_j) = \alpha_j$ **simultaneous Diophantine**: there exists $c, \tau > 0$ s.t.

$$\inf_{l \in \mathbb{Z}} \max_j |n\alpha_j - l| \geq \frac{c}{|n|^\tau}, \quad \forall n \in \mathbb{Z} \setminus \{0\}$$

$\implies$ Local C^∞ simultaneous linearization

► **Fayad-Khanin 2009**: Global C^∞ simultaneous linearization under the same assumption as **Moser 1990**

Question: Rigidity of Translations on the Torus

Fayad-Krikorian ICM 2018: Let $F : \mathbb{T}^d \rightarrow \mathbb{T}^d$ be a smooth diffeomorphism of the torus $\mathbb{T}^d$ which is topologically conjugate to a translation $R_\alpha : \mathbb{T}^d \rightarrow \mathbb{T}^d$, $x \mapsto x + \alpha$ with $\alpha \in \mathbb{T}^d$ **Diophantine**. Is the conjugacy smooth?

Conjecture: Rigidity in More General Situations

Khanin ICM 2018: Let M be a compact Riemannian manifold. Let T_1 and T_2 be C^∞ diffeomorphisms of M . Assume also that T_1 and T_2 satisfy the **Diophantine property**. Namely, there exists $\tau, C > 0$ such that

$$\text{dist}(T_1^n x, x), \text{dist}(T_2^n x, x) \geq Cn^{-\tau}, \quad \forall x \in M, \quad \forall n \in \mathbb{N}. \quad (1)$$

Rigidity Conjecture. Suppose T_1 and T_2 are topologically conjugate. Then the conjugacy is C^∞ —smooth. Probably it is enough to require the Diophantine condition (1) for only one of the maps T_1 and T_2 .

For general compact Riemannian manifold,

- No rotation number
- Arithmetic conditions in a more general form ?
- Methodology for the circle diffeomorphism could fail

Affirmative Answers for Concrete Manifold

Dolgopyat-Krikorian 2006: “Zero Lyapunov exponent of random walk”
 $\iff$ Local C^∞ simultaneous linearization of diffeomorphisms of $\mathbb{S}^d$

Karaliolios 2018: $\alpha \in \mathbb{T}^d$ Diophantine, α is in the convex hull of the rotation set of F
 $\implies$ Local C^∞ linearization of diffeomorphisms of $\mathbb{T}^d$

Petković 2021: $\{\alpha_j\} \in \mathbb{T}^d$ simultaneous Diophantine, α_j is in the convex hull of the rotation set of F_j
 $\implies$ Local C^∞ simultaneous linearization of finitely many commuting diffeomorphisms of $\mathbb{T}^d$

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Counter-example to Rigidity Conjecture on $\mathbb{T}^2$

Forni-Kanigowski 2021: There exist C^∞ diffeomorphisms T_1 and T_2 of $\mathbb{T}^2$, which satisfy the Diophantine condition (1), are C^0 conjugate but not C^1 conjugate.

- **Arnold 1961:** There exists Liouville irrational number β satisfying: There exists a C^∞ , or even real-analytic, circle diffeomorphism $g : \mathbb{T} \rightarrow \mathbb{T}$ topologically conjugate to R_β such that the conjugacy is not of class C^1 .

$\rightsquigarrow$ For α **Diophantine**, $T_1 = R_\alpha \times R_\beta$ and $T_2 = R_\alpha \times g$

Group Action by Isometries π

Finitely presented group Γ :

- Generators and their inverses $G = \{\gamma_1, \dots, \gamma_k\}$
- Relations $R = \{\mathcal{W}_1, \dots, \mathcal{W}_p\}$: each $\mathcal{W}_i$ is a finite word of elements in G equal to e , i.e., for $j = 1, \dots, p$,

$$e = \mathcal{W}_j = \gamma_{l_1^{(j)}} \cdots \gamma_{l_{m_j}^{(j)}}, \quad 1 \leq l_1^{(j)}, \dots, l_{m_j}^{(j)} \leq k.$$

EXAMPLE. For the abelian group, $R = \left\{ \gamma_i^{-1} \gamma_j^{-1} \gamma_i \gamma_j : 1 \leq i, j \leq \frac{k}{2} \right\}$

Let (M, g) be a real-analytic compact Riemannian manifold. Define the Γ -action by **analytic isometries** $\pi : \Gamma \rightarrow \text{Isom}(M, g) \subset \text{Diff}^\omega(M)$, i.e., for $\gamma \in \Gamma$, $\pi(\gamma)$ is an analytic diffeomorphism of M which preserves the distance induced by the Riemannian metric g .

Hochschild Complex of Cochains of L^2 Vector Fields

- **Hochschild complex** of cochains of L^2 vector fields:

$$C_0(\Gamma, L^2(M, TM)) \xrightarrow{d_0} C_1(\Gamma, L^2(M, TM)) \xrightarrow{d_1} C_2(\Gamma, L^2(M, TM)) \xrightarrow{d_2} \dots,$$

with the homomorphisms satisfying $d_{n+1} \circ d_n = 0, \forall n$

- ▶ $C_0(\Gamma, L^2(M, TM)) = L^2(M, TM)$
 - ▶ $C_1(\Gamma, L^2(M, TM)) = (L^2(M, TM))^k$
 - ▶ $C_2(\Gamma, L^2(M, TM)) = (L^2(M, TM))^p$
- **Cohomology groups** $H^n(\Gamma) := \text{Ker} d_{n+1} / \text{Im} d_n$

Homomorphisms d_0 and d_1

- ▶ For $v \in L^2(M, TM)$, $d_0 v = (v - \pi(\gamma_l)_* v)_{1 \leq l \leq k}$.
- ▶ For $V \in (L^2(M, TM))^k$ and the finite words $\mathcal{W}_j = \gamma_{l_1^{(j)}} \cdots \gamma_{l_{m_j}^{(j)}} \in R$,

$$d_1 V = \left(\sum_{1 \leq z \leq m_j} \left(D\pi \left(\prod_{i=1}^{z-1} \gamma_{l_i^{(j)}} \right) \cdot V_{l_z^{(j)}} \right) \circ \left(\prod_{i=z}^{m_j} \pi \left(\gamma_{l_i^{(j)}} \right) \right) \right)_{1 \leq j \leq p}.$$

EXAMPLE. For rotations $R_{\alpha_j} : x \mapsto x + \alpha_j$ on $M = \mathbb{T}$ (hence commuting)

- ▶ $(d_0 v)(x) = (v(x) - v(x \pm \alpha_j))_j$
- ▶ $(d_1 V)(x) = (V_i(x + \alpha_j) - V_i(x) - V_j(x + \alpha_i) + V_j(x))_{i,j}$

Self-adjoint Box operator

- Following Moser 1990 and Lombardi-Stolovitch 2010, define the self-adjoint box operator

$$\begin{aligned}\square : (L^2(M, TM))^k &\rightarrow (L^2(M, TM))^k \\ V &\mapsto \square V := (d_0 \circ d_0^* + d_1^* \circ d_1)V .\end{aligned}$$

where d_0^* and d_1^* denote the adjoints of d_0 and d_1 respectively, with respect to the scalar product from $L^2(M, TM)$ induced on $(L^2(M, TM))^k$ and $(L^2(M, TM))^p$ respectively.

- Related to the first cohomology: $\text{Ker} \square = H^1(\Gamma) = \text{Ker} d_1 / \text{Im} d_0$.

Diophantine Condition of Group Action Isometries π

$\{\lambda_j\}_{j \in \mathbb{N}}$ – the eigenvalues of $|\Delta_{TM}|^{\frac{1}{2}}$, with $0 = \lambda_0 < \lambda_1 < \dots \rightarrow \infty$

Zimmer 1990: Peter-Weyl decomposition $L^2(M, TM) = \bigoplus_{n \geq 0} E_{\lambda_n}$

μ_n – the smallest non-zero eigenvalue of $\square_n = \square \circ \mathbb{P}_{E_{\lambda_n}^k}$ (if existing)

Definition (Similar to Definition of Dolgopyat 2002)

The Γ –action by isometries π is called **Diophantine** if there exists non-negative numbers σ, τ such that the operator $\square$ satisfies

$$\mu_n \geq \frac{\sigma}{(1 + \lambda_n)^\tau}, \quad \forall n \in \mathbb{N}. \quad (2)$$

EXAMPLE. For finitely many rotations $R_{\alpha_j} : x \mapsto x + \alpha_j$ on $\mathbb{T}$, $\lambda_n = n$

Moser 1990: $\mu_n = 4 \sum_j \sin^2(\pi n \alpha_j)$

$\{\alpha_j\}$ **simultaneous Diophantine** $\implies$ Diophantine condition (2) of action by rotations

Formal conjugacy

Consider a Γ -action by analytic diffeomorphisms $\pi_0 : \Gamma \rightarrow \text{Diff}^\omega(M)$, which is a **perturbation of π** , written as:

$$\pi_0(\gamma) = \text{Exp}\{P_0(\gamma)\} \circ \pi(\gamma), \quad \gamma \in G,$$

where Exp is the exponential map defined upon the Riemannian connection ([Helgason 1978](#)).

Definition (Formal conjugacy)

Given $0 < \zeta < 1$, π_0 is said to be **ζ -formally conjugate** to π on M , if, for any $\varepsilon > 0$, there exists $y^\varepsilon \in \text{Vect}^\omega(M, TM)$ with $\|y^\varepsilon\|_{C^1, M} < \zeta$ such that

$$\text{Exp}\{y^\varepsilon\}^{-1} \circ \pi_0(\gamma) \circ \text{Exp}\{y^\varepsilon\} = \text{Exp}\{z^\varepsilon(\gamma)\} \circ \pi(\gamma), \quad \forall \gamma \in G,$$

for some $z^\varepsilon : \Gamma \rightarrow \text{Vect}^\omega(M, TM)$ satisfying $\|z^\varepsilon\|_{G, C^1, M} < \varepsilon$.

Main Theorem 1

For the Γ -action by analytic diffeomorphisms π_0 on M , with

$$\pi_0(\gamma) = \text{Exp}\{P_0(\gamma)\} \circ \pi(\gamma), \quad \gamma \in G,$$

where $P_0 : G \rightarrow H_{r_0}^2(M, TM)$ (**Hardy space**) for some sufficiently small r_0 and with small

$$\|P_0\|_{G, H_{r_0}^2} = \sum_{\gamma \in G} \|P_0(\gamma)\|_{H_{r_0}^2} =: \varepsilon_0$$

Theorem (Stolovitch-Z. 2023)

Assume that π is **Diophantine** and satisfies $\dim \text{Ker} \square < \infty$. If π_0 is ζ -**formally conjugate** to π on M with $0 < \zeta < |\ln(\varepsilon_0)|^{-\frac{1}{3}}$, then π_0 is analytically conjugate to π .

$\implies$ Analytic version of generalized Rigidity Theorem of Moser 1990 and Petković 2021

Main Theorem 2

For the Γ -action by analytic diffeomorphisms π_0 on M , with

$$\pi_0(\gamma) = \text{Exp}\{P_0(\gamma)\} \circ \pi(\gamma), \quad \gamma \in G,$$

where $P_0 : G \rightarrow H_{r_0}^2(M, TM)$ for some sufficiently small r_0 and with small

$$\|P_0\|_{G, H_{r_0}^2} = \sum_{\gamma \in G} \|P_0(\gamma)\|_{H_{r_0}^2} =: \varepsilon_0$$

Theorem (Stolovitch-Z. 2023)

Assume that π is *Diophantine* and satisfies $H^1(\Gamma) = 0$. Then π_0 is analytically conjugate to π .

$\rightsquigarrow$ “Analytic version” of Rigidity Theorem of [D. Fisher 2007](#)

Idea of proof

KAM scheme for the Γ -action $\pi_0 = \text{Exp}\{P_0\} \circ \pi$: for $r_j \searrow \frac{r_0}{2}$

$$\begin{array}{lll}
 \text{Exp}\{P_0\} \circ \pi & P_0 : \Gamma \rightarrow H_{r_0}^2(M, TM) & \|P_0\|_{G, H_{r_0}^2} = \varepsilon_0 \\
 y_0 \downarrow & y_0 \in H_{r_1}^2(M, TM) & \\
 \text{Exp}\{P_1\} \circ \pi & P_1 : \Gamma \rightarrow H_{r_1}^2(M, TM) & \|P_1\|_{G, H_{r_1}^2} < \varepsilon_0^{\frac{5}{4}} \\
 \downarrow & & \\
 \vdots & & \\
 \downarrow & & \\
 \text{Exp}\{P_j\} \circ \pi & P_j : \Gamma \rightarrow H_{r_j}^2(M, TM) & \|P_j\|_{G, H_{r_j}^2} < \varepsilon_0^{(\frac{5}{4})^j} \\
 y_j \downarrow & y_j \in H_{r_{j+1}}^2(M, TM) & \\
 \text{Exp}\{P_{j+1}\} \circ \pi & P_{j+1} : \Gamma \rightarrow H_{r_{j+1}}^2(M, TM) & \|P_{j+1}\|_{G, H_{r_{j+1}}^2} < \varepsilon_0^{(\frac{5}{4})^{j+1}} \rightarrow 0 \\
 \downarrow & & \\
 \vdots & & \\
 \pi & &
 \end{array}$$

The j^{th} -KAM step

Aim: To eliminate the terms of

$$(P_j(\gamma))_{\gamma \in G} \in (H_{r_j}^2(M, TM))^k \subset (L^2(M, TM))^k = \text{Ker} \square \oplus \text{Im} \square$$

► Diophantine condition of $\square$ (hence π)

$\rightsquigarrow$ Elimination of terms in $\text{Im} \square$

► Formal conjugacy & $\dim \text{Ker} \square < \infty$ or $H^1(\Gamma) = 0$

$\rightsquigarrow$ “Elimination” of terms in $\text{Ker} \square$

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Thanks for your attention!